

Analysis of BRS Document

PART-I (CMMS)

#	Ref. Text of BRS Doc	Page No. of BRS Doc	C-DAC Remarks	Remarks of ORGI (As on 02.04.25)
1	1.2.1 Provision for opting NPR to be excluded or included with HLO will be worked out at the time of design and development of CMMS.	5	Decision needs to be confirmed during the requirement analysis phase.	CMMS for HHC may be done separately independent of NPR. NPR part in CMMS, NPR App and SE portal for NPR may be prepared separately at later stage as per the requirement.
2	1.2.8 Additionally, another 1.5 lakh other Census functionaries will receive training from ORGI officials, who may themselves be trained by NTs or Subject Matter Experts at State/District/Sub-District Level.	8	This is the new requirement and not covered under the current scope of work.	The training system of other census functionaries will be kept outside CMMS. No need to develop any separate module for Census functionaries other than NT/MT/FT and E&S.
3	1.2.8 Attendance of Trainers	8	This needs to be discussed in detail.	Trainer's attendance will be covered in offline mode. The trainees will also put their attendance on the physical attendance sheet.
4	2.1 b e) Integration of Geo-location of HLBs	9	This is the new requirement and not covered under the current scope of work.	<i>Page 12 of Proposal: As discussed in the meeting with C-DAC held on 18.03.2025 at Shastri park in presence of IT Division.</i> Requirement 1: <i>E/S User Info for HLB/ Digital Layout Mapping Apps from CMMS to be securely shared for authorising access to Mapping Apps (HLB Mobile Mapping App and Digital Layout Mapping App).</i>

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				<i>The required fields are mentioned in the BRS document.</i> Requirement 2: <i>Boundary and Geo-Location of landmark (if available) of HLBs/EBs</i> <i>To locate HLBs/EBs by E/S, the coordinates of the assigned HLBs/EBs to be shared through HLO/PE Apps.</i> <i>The required coordinates for HLBs/EBs are to be sourced either from existing Census 2011 EBs or using Web Mapping Module for capturing HLB/EB coordinates to be facilitated within CMMS for Charge User access.</i> <i>The development of this module to be done by ESRI Team. IT/Census/Map Divisions along with C-DAC and ESRI to discuss about Tech Stack to be used and other operational aspects of the module integration.</i> <i>The enumerator will prepare HLB/EB digital layout map through the DLM App to be prepared by ESRI. The number of buildings, census houses, households obtained during DLM activity will be autofilled through CMMS in the HLO data collection app. If the information is not available through DLM then the enumerator will physically insert the number of</i>

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				<i>census houses in the HLO app.</i>
5	2.1 c 6.1 Data Entry of data collected manually by Enumerators (Post Enumeration Surveys)	9	This is the new requirement and not covered under the current scope of work.	Development of the same will be taken up afterwards and decisions conveyed accordingly
6	2.1 m 5) Charge Officer will arrange to get the data entry completed from the corrected booklets on the CMMS portal.	17	Who will do the data entry, Charge Officer or DEO?	Data entry may be done through Enumerator's login (as decided in EFC Meeting).
7	2.1 n 3 c When existing Census 2011 EBs are split, or additional block formations occur, the provision to add location coordinates for such blocks must be facilitated. Map Division will coordinate the development requirement in this regard between C-DAC and ESRI teams.	18	This should be finalized during the requirement analysis phase.	Frame of 2011 EB may not be used on CMMS portal for the creation of HLBs/EBs for development of CMMS. However, the Web Mapping Module for capturing HLB/EB coordinates mentioned against Query No. 4 can be used for this purpose also. It was clarified that whether Census 2011 EB are used for reference for creation of fresh HLBs for the ensuing Census, the provision of geotagged co-ordinates should be there. Map Div. will be responsible for providing the location coordinates for their integration with the CMMS portal.
8	2.1 n 3 Note: Availability of Census 2011 EB maps in georeferenced form depends on the coverage / completion of Census 2011 EB mapping which is in progress now. Format for HLB/EB to be decided later.	18	This should be finalized during the requirement analysis phase.	Frame of 2011 EB may not be used on CMMS portal for the creation of HLBs/EBs for development of CMMS.
9	2.1 o 3 a Knowledge Transfer Session	19	This should be finalized during the requirement	<i>Meeting with C-DAC team was held on 18.03.2025 at Shastri park. C-DAC team was appraised on</i>

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			analysis phase.	<i>MAP production Environment.</i>
10	4.1 Security & Privacy - Role-Based Authorization	29	What are the different roles in case of Self-Enumeration Web Portals?	This was generic and was meant to authorise correct. Refer Page 15-17 of Proposal.
11	4.2 Any member of the household can do Self-enumeration (SE) but only once for the entire household. In case more than one member of a household has self-enumerated for the household, the Reference ID returned from the field will be valid.	29	This needs to be discussed in detail.	Explanation: If more than one member of the household has done SE then only one Reference ID will be collected from the field as given by respondent and noted. The other IDs will be discarded later on.
12	4.3 Login Facility: At PE phase provision should be available for households to self-enumerate even if mobile number is not given in first phase.	31	This is the new requirement and not covered under the current scope of work.	This is a part of login process & is required to facilitate SE for the households who have not returned mobile number at the time of HLO & facility need to be provided in the system.
13	4.4 b) User be allowed to login to portal after authentication using credentials like mobile no., Aadhaar no., demographic information stored in NPR database.	33	This is the new requirement and not covered under the current scope of work.	SE portal for NPR may be prepared separately at later stage if requirement is provided for the same.
14	4.4 d) Separate provision for Registration of login particular in NPR database and confirmation/ objection to Registration.	33	This needs to be discussed in detail.	
15	4.4 j) New households may also be added.	33	This needs to be discussed in detail.	
16	5.1 ChatBot will support 16 languages	35	This is the new requirement and not covered under the current scope of work.	Refer Page No 17 of proposal: <i>Creation and Integration of Chatbot with CMMS/Website</i>

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				No particular language was mentioned in the proposal. But ChatBot is required in as many languages as technically possible.
17	5.1 Objectives: Ensure accessibility across Web, Mobile, IVR, and WhatsApp.	35	This is the new requirement and not covered under the current scope of work.	These are part of third party integration as given in Page 14 of proposal. However, IVR may not be required if technically difficult. C-DAC informed that they will evaluate the requirements related to WhatsApp.
18	5.1 Key Requirements: voice-based queries	35	This is the new requirement and not covered under the current scope of work.	No particular feature of Chatbot was mentioned in the proposal. This feature is required. C-DAC will further explore issues related to Chatbot.
19	5.1 Key Requirements: role-based access	35	This is the new requirement and not covered under the current scope of work.	No process of Chatbot accessibility was mentioned in the proposal. This feature is required.
20	5.1 Key Requirements: Machine learning-based self-improvement	35	This is the new requirement and not covered under the current scope of work.	No particular feature of Chatbot was mentioned in the proposal. This feature is required.
21	6.1 PES App – HLO & PE	37	This is the new requirement and not covered under the current scope of work.	Included in the proposal at page 10 as “Mobile applications for Post Enumeration Surveys (PES)”
22	6.2 Product 11 - Mobile App and Portal for Training	37	This is the new requirement and not covered under the current scope of work.	Included in the proposal at page 10 as “Module for Training of Census functionaries on CMMS & Data Entry Apps in Android and iOS through Container App”.

PART- II (Mobile App)

S. No.	Page no. Reference	BRS Description	Mobile App team Remarks	Remarks of ORGI (As on 02.04.25)
1.	Page 15 h.9	User management-> Provision of verification of mobile number and validation of user credentials. Verification of mobile number by enumerator should be compulsory before field work. However, validation for enumerator credentials should not be mandatory before the start of field operation. It should be mandatory before completion of field work.	This should be verified before starting the field work, through OTP verification at the time of mobile app login	As per BRS, verification of registered mobile number will be done mandatorily through OTP before the field work but validation of enumerators credentials in CMMS may be done afterwards but before completion of field work.
2.	Page 17 m. data entry->2	HLO Enumerator Abstracts by CMMS / by supervisor on mobile	Abstract seeding: As per our understanding, Abstract seeding is digitization of 'summary/total' fields of paper mode schedule which are submitted by the enumerator. • It is not feasible to do this from the app by a supervisor. • Paper-based entry will be digitized through CMMS portal as informed earlier.	For the paper mode data collection, enumerator abstract need to be entered by the supervisor in the supervisor module. Already explained to C-DAC.
3.	Page 18-> Maps of census 2011 EBs	After the re-serialisation process, georeferenced Census 2011 EB maps must be provided to all E&S through the HLO App for field use.	The format to be provided by ORGI and what additional information will be given regarding HLB?	It was clarified that whether Census 2011 EB are used for reference for creation of fresh HLBs for the ensuing Census, the provision of geotagged co-ordinates should be there. Map Div. will be responsible for

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				providing the location coordinates for their integration with the CMMS portal.
4.	Page 21 - > 3.1 -> secured login	Single Sign On (SSO) in Apps	Is it using the same credentials for all three apps? Need further discussion.	Yes, interface for single sign on. Container app may facilitate this.
5.	Page 21 - > 3.1 -> Real-Time Data Update & Syncing	Real-Time Data Update & Syncing	This needs to be discussed in detail.	Real time indicates that the time of data synchronized by the enumerators. PMG 2 & C-DAC may discuss further.
6.	Page 21 - > 3.1 -> In-App Help	In-App Help	Who will provide content for 'Help & FAQ'?	ORGI
7.	Page 22 - > 3.2	Data capturing-> Integration with mapping mobile application and No. of Census House as per Layout Map should be available in HHC app	This is the new requirement and not covered under the current scope of work.	<i>Meeting with C-DAC held on 18.03.2025 at Shastri park.</i> <i>Integration with MAP Prod is limited to updating information on No. of Census House or Households against each HLB/EB through back-end update procedure.</i>
8.	Page 22 - > 3.2	Data capturing-> Also provision to record No. of Census House (CH) as per Layout Map (de-novo) should be there.	This is the new requirement and not covered under the current scope of work.	<i>As above</i>

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9.	Page 22-> 3.2	HLB Completion Conditions -> Case (i): If canvassed Census Houses match the recorded number, HLB can be marked complete.	Not clear to us, needs further discussion.	If the number of canvassed Census houses is less than the number recorded at the beginning the HLB will not be marked as complete. If practically the situation is so then the enumerator has to change the number of total census houses entered at the beginning, then only can show HLB as completed. If the number of canvassed census houses is more than the number recorded at the beginning, then a pop up will come that number of Census Houses is exceeding the number given in the beginning. Do you want to continue? If yes is entered, then he will be able to continue the work.
10.	Page 22-> 3.2	Self-enumeration ID recording-> Mobile number should be disabled and Summary of household up to Q10 along with reference ID should be shown.	Which mobile number and when to disable mobile number?	After recording the SE Reference ID Q No 11 to Q34 to be disabled and the respondent will only be shown the Summary of household up to Q10 along with reference ID.
11	Page 23->	Special Provisions	This is the new requirement and not	This is part of Process for

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	3.2		covered under the current scope of work.	validation checks. Extra information was provided for the benefit of C-DAC
12	Page 23-> 3.2	Summary & Editing Features-> - Summary screen of all filled responses at the end of canvassing a household.	It was informed earlier that the summary will be shown in the CMMS portal. Need further discussion.	This is a part of the process. It will be shown in the Mobile App only for each enumerator for each household for his/her EB.
13	Page 23-> 3.2	Summary & Editing Features-> Delete option for specific Census House records.	This is to be done before or after the sync? Need further discussion.	Before Sync.
14	Page 24-> 3.2	Edit after sync option-> Reason for editing already synced data should be recorded (as per the drop-down – competent census functionaries).	Need further discussion.	The reason (as observed by self/supervisor/Charge officer/State Officer/ORGI officer/DCO officer etc.) for editing after sync will be given in the drop down menu.
15	Page 24 - >3.2	Supervisor Module -> Provision of abstract seeding for paper mode enumerator should be there. Provision for capturing SE id if done on paper mode.	This is the new requirement and not covered under the current scope of work.	Already included in the proposal at Page No 13: <i>Abstract Seeding module for Supervisor</i> . Checking the correctness of SE ID if done on paper schedule will be provided in Abstract Seeding module
16	Page 25-> 3.3	Census House Enumeration -> Enumerators must select a Census House (prefilled from HLO) from the dropdown list, fetching details from HLO.	Need further discussion.	This is to be done to link HLO & PE
17	Page 25-	Present use of Census House	Need further discussion.	Please see page 27 for further

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	>3.3			explanation
18	Page 25->3.3	Self-enumeration ID recording in PE	Need further discussion.	After selecting a present use as residence, residence-cum-other use a message need to be displayed saying “Has the Self-enumeration been completed by the household? With options Yes/No.
19	Page 25 -> 3.3	Editing & Signature-> Option to delete a specific CH/HH record if necessary.	On what basis delete action will take place? Can it be done after sync also?	In case of error by enumerator in entering data. No only before sync.
20	Page 25 -> 3.3	Provision of Signature-> Functionality to capture and attach images (of signature) at all the time of uploading data	It should be jpg or finger sign will be enough. Please clarify.	Finger sign
21	Page 25 -> 3.3	Supervisor Module ->Provision for capturing SE id if done on paper mode.	This is the new requirement and not covered under the current scope of work.	Checking the correctness of SE ID if done on paper schedule will be provided in Abstract Seeding module
22	Page 27	* For New Household/Census House, data collection after selecting present use will continue as follows:	Need further discussion.	Already explained in point no 17
23	Page 28	3.4 -> 18 languages	In other parts of the app it is in 16 languages	It’s actually 16 languages. Typo regretted.

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24	Page 28	The field work for NPR data collection was undertaken along with the House listing and Housing Census 2010. NPR Database has been updated during 2015-16 in all States/UTs (except Assam and Meghalaya) to make a comprehensive resident database	Need further discussion.	This is a statement
25		Product 9,10,11	This is the new requirement and not covered under the current scope of work.	The matter of Product 9 (Container App) has been discussed in number of meetings and also mentioned in page 10 of the proposal. Product 10 & 11 were there in original proposal.
26	Page 38	Product 12-> manuals in audio visual format	This is the new requirement and not covered under the current scope of work.	Audio-Visual operating manual is required for easy understanding of the users.

ORGI further comment:

- Development of CMMS and availability for testing should be done module wise.
- SRS for individual applications & portal should be prepared separately so that development of modules may be started in parallel.

Other Questions:

- 1- Need more clarity for NPR mobile app like edit functionality, SE ID capturing, search and edit etc.
- 2- What all details will be fetched from AHL in PE app?
- 3- Abstract seeding and SE ID capturing in Supervisor role needs more clarity.
- 4- All details related to PES should be provided like process of allotment, questionnaire, help documents etc.
- 5- Data about location particulars and HLBS

--- **ORGI Reply** : This has been discussed in details and informed C-DAC